

Motherboard

The right base

The scenario in the motherboard market has radically changed over the past year or so. There are now choices for every single type of consumer need, beginning from 'plain jane' motherboards, to fully featured powerhouses, for every single type of mainstream processor. However, technology changes so quickly that yesterday's powerhouses become out-dated tomorrow. Owning the 'best there is', is a short lived pleasure. Thus, keeping up with the technology curve for motherboards could mean an investment of thousands of rupees each year.

There are ways to avoid this constant upheaval of hardware, and it begins by choosing the right technology and feature set. This alone can ensure that your PC is ready to tackle the next-generation of hardware.

Jargon busters

The primary function of a motherboard is to provide a base upon which other components, such as the processor and RAM, can reside. Since different types of components reside on the motherboard, it also has to be capable of co-ordinating the flow of information between these components. It also orchestrates the transfer of data and information within the system.

Smart tips

Processor interface: This is the socket on the motherboard that the processor is mounted on.

There are generally three types of slots available: Socket 370 (for Via), Socket A and Socket 478.

Chipset: Different chipsets are used to support functions that are offered by the board—the RAM and, more importantly, the processor. For example, the Intel 845 chipset supports Pentium 4 and DDR SDRAM. AMD processors (Duron and Athlon) require VIA KT 333 or 400 chipset-based motherboards.

Interface slots: These slots are used to connect add-on cards such as a soundcard or a network card, to the computer.

There are primarily two types of slots found on today's motherboards—PCI and AGP. PCI cards are the most commonly available, whereas the AGP interface is meant solely for high-speed graphics cards.

RAM slots: DDR RAM is the most popular option today, with SD RAM fading into oblivion. Most motherboards have at least two slots, but more expensive models have four slots for RAM. Remember that the total amount of RAM that you can install in your system is also dependant on your motherboard chipset.

BIOS: Every motherboard carries a special circuitry called the BIOS (Basic Input Output System), which exists on a Read Only Memory (ROM) chip. The 'jumperless' motherboards that are the norm today, allow all parameters to be automatically detected and configured. For example, the processor core voltage and clock multiplier, and the configuration and control of integrated peripherals are all auto-detected.

Also Remember...

■ Always check the maximum CPU speed supported.

■ Look for onboard USB ports, options for additional USB ports and also for Firewire ports.

■ Onboard thermal and fan speed monitoring are a must for servers and AMD users.

■ Voltage and Bus speed tweaking, for overclocking is an added advantage.

■ Buy an ATX or Micro power supply—the ATX power supply has better power management features.

■ The higher the number of IDE devices supported, the better.

■ Look for support for ATA-133 IDE, Serial ATA standards.

■ Check for the highest FSB (Front Side Bus) speed supported.

■ Support for ECC (Error Correction Capability) RAM is advisable.

■ Integrated features, such as onboard soundcard and display card (for SoHo users) will save you a pretty penny.

■ Look for additional features, such as Wake-on-Modem-ring (integrated into the BIOS)—it allows the computer to power on automatically for receiving faxes.

Decision Maker

	Bare necessities	Future resistant	Performance seeker
You need	Basic support for current processors and peripherals	Flexibility and upgradeability	Maximum performance and overclockability.
Look for	Minimum three PCI slots with integrated sound and video in the Micro-ATX format	Minimum five PCI slots, three RAM slots, an AGP 4x slot in the ATX format.	The latest chipset for your processor and options for easy over clocking
Our pick	Mercury KOB650GL NDSMx (Intel), ASUS A7S266 (AMD)	DFI NT72-SA (Intel), ASUS A7S266 (AMD)	MSI 845PE Max2 (Intel), MSI KT4 Ultra (AMD)
Price range	Rs 3,000 to Rs 5,000	Rs 5,000 to Rs 8,000	Rs 10,000 and above